LAPORAN FINAL PROJECT

MOCHI: MINECRAFT OBJECT COMPREHENSION & HYBRID INDEXING

FAIZ MUHAMMAD KAUTSAR

NRP 5054231013

EFAN RAMDHANI

NRP 5054231017

MUHAMMAD RASYAD LUBIS

NRP 5054231010

MUHAMMAD FARHAN ARYA WICAKSONO

NRP 5054231011

Dosen Pengampu

Dosen Pengampu 1

NIP 1xxxxxxxxxxxxxxxx

Dosen Pengampu 2

NIP 1xxxxxxxxxxxxxxxx

Program Studi Rekayasa kecerdasan artifisial

Departemen Teknik informatika

Fakultas Fakultas Teknologi Elektro dan Informatika Cerdas

Institut Teknologi Sepuluh Nopember

Surabaya

Tahun 2025

LAPORAN FINAL PROJECT

MOCHI: MINECRAFT OBJECT COMPREHENSION & HYBRID INDEXING

FAIZ MUHAMMAD KAUTSAR

NRP 5054231013

EFAN RAMDHANI

NRP 5054231017

MUHAMMAD RASYAD LUBIS

NRP 5054231010

MUHAMMAD FARHAN ARYA WICAKSONO

NRP 5054231011

Dosen Pengampu

Dosen Pengampu 1

NIP 1xxxxxxxxxxxxxxxx

Dosen Pengampu 2

NIP 1xxxxxxxxxxxxxxxx

Program Studi Rekayasa kecerdasan artifisial

Departemen Teknik informatika

Fakultas Fakultas Teknologi Elektro dan Informatika Cerdas

Institut Teknologi Sepuluh Nopember

Surabaya

Tahun 2025

ABSTRAK

MOCHI: MINECRAFT OBJECT COMPREHENSION & HYBRID INDEXING

Nama Mahasiswa / NRP	: Faiz Muhammad Kautsar / 5054231013 Efan Ramdhani / 5054231017 Muhammad Rasyad Lubis / 5054231010 Muhammad Farhan Arya Wicaksono / 5054231011
Departemen	: Teknik informatika
Dosen Pembimbing	: Dosen Pengampu 1

Abstrak

Minecraft hosts one of the largest user-generated 3D content ecosystems, yet searching for community-created schematics relies on inconsistent, keyword-based tags. We present MOCHI, a cross-modal retrieval system between natural language and 3D voxel structures in the Minecraft domain. By leveraging a CLIP-style dual-encoder architecture with a 3D CNN and a frozen language model, we learn a shared embedding space for text and discrete voxel grids. We introduce a two-stage training pipeline incorporating Masked Voxel Modeling (MVM) for self-supervised pretraining, followed by symmetric InfoNCE contrastive fine-tuning. We demonstrate that bounding-box cropping and MVM pretraining significantly improve retrieval performance on a dataset of 8,328 community schematics, achieving a top-1 category hit rate of over 50%.

Kata kunci: *kata kunci 1, kata kunci 2*

ABSTRACT

MOCHI: MINECRAFT OBJECT COMPREHENSION & HYBRID INDEXING

Student Name / NRP	: Faiz Muhammad Kautsar / 5054231013 Efan Ramdhani / 5054231017 Muhammad Rasyad Lubis / 5054231010 Muhammad Farhan Arya Wicaksono / 5054231011
Department	: Teknik informatika
Advisor	: Dosen Pengampu 1

Abstract

Fill in the English abstract here.

Keywords: *keyword 1, keyword 2*

DAFTAR ISI

DAFTAR GAMBAR

DAFTAR TABEL

DAFTAR SIMBOL

Simbol	Keterangan

BAB 1 PENDAHULUAN

1.1 Latar Belakang

Minecraft is home to one of the largest user-generated 3D content ecosystems in the world. Community platforms host hundreds of thousands of player-created structures, shared as downloadable schematics (3D voxel grids of block IDs). However, finding the right schematic is painful as search is predominantly keyword-based, tags are inconsistent, and there is no way to search by structure or semantic intent.

This paper explores cross-modal retrieval between natural language and 3D voxel structures in the Minecraft domain. We learn a shared embedding space where text descriptions and 3D block grids live side-by-side, enabling bidirectional semantic search. Our approach draws from the CLIP paradigm, utilizing a dual-encoder architecture trained with symmetric InfoNCE loss. It is adapted for discrete voxel grids and community-authored text, using a relatively small dataset of approximately 8,000 samples.

(Ruan et al., 2024)

1.2 Rumusan Masalah

Permasalahan penelitian harus dituliskan dalam bentuk deklaratif atau kalimat-kalimat pertanyaan yang tegas dan jelas.

1.3 Batasan Masalah

Ruang lingkup/pembatasan masalah dalam upaya memfokuskan penelitian yang akan dilakukan menjadi lebih terarah.

1.4 Tujuan

Tujuan penelitian/tugas akhir/desain.

1.5 Manfaat

Manfaat penelitian/tugas akhir/desain.

BAB 2 TINJAUAN PUSTAKA

2.1 Hasil Penelitian Terdahulu

2.2 Dasar Teori

BAB 3 METODOLOGI

3.1 Metode yang Digunakan

3.2 Bahan dan Peralatan yang Digunakan

3.3 Urutan Pelaksanaan Penelitian

BAB 4 HASIL DAN PEMBAHASAN

BAB 5 KESIMPULAN

DAFTAR PUSTAKA

Ruan, Y., Lee, H.-H., Zhang, Y., Zhang, K., & Chang, A. X. (2024). Tricolo: Trimodal contrastive loss for text to shape retrieval. *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 5815–5825.

LAMPIRAN